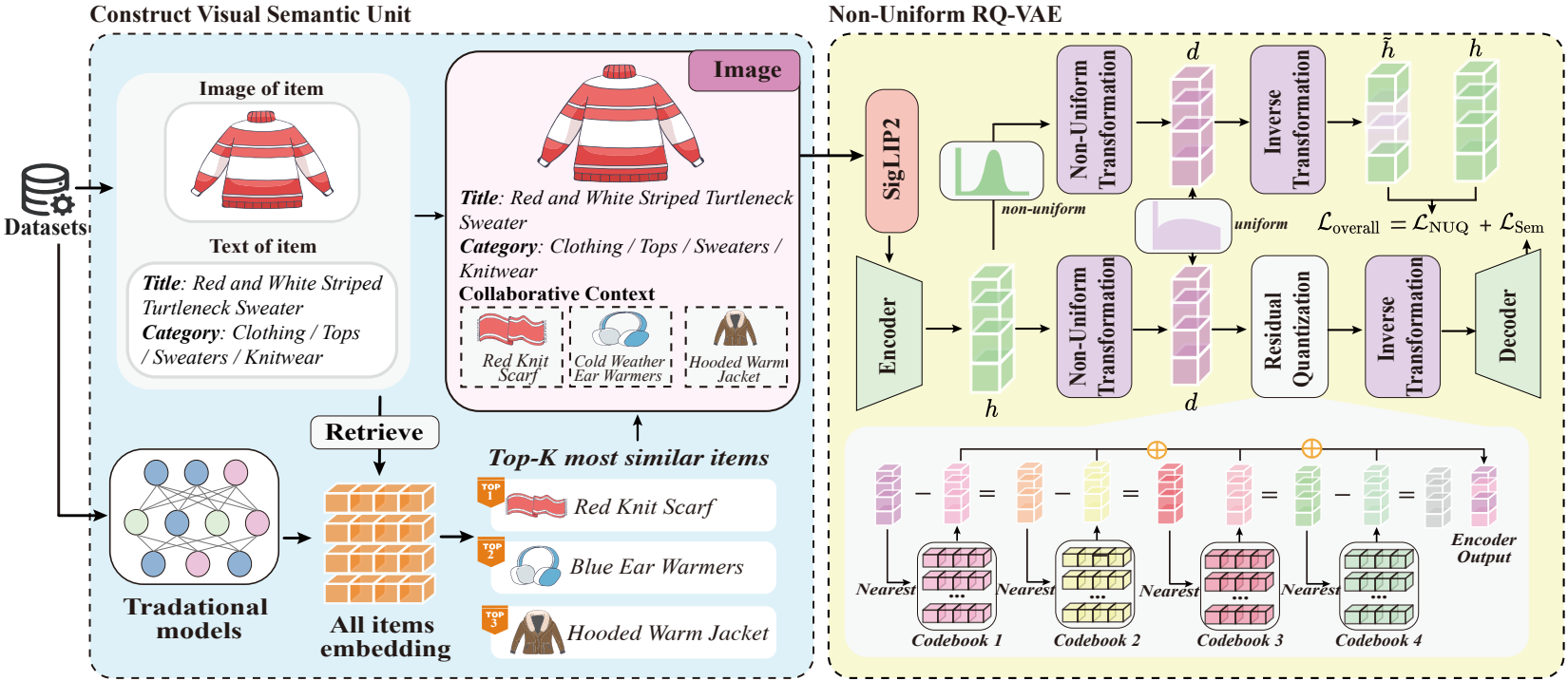
Construct Visual Semantic Unit
Non-Uniform RQ-VAE
Datasets
Image of item
Text of item
Title: Red and White Striped Turtleneck Sweater
Category: Clothing / Tops / Sweaters / Knitwear
Image
Title: Red and White Striped Turtleneck Sweater
Category: Clothing / Tops / Sweaters / Knitwear
Collaborative Context
Red Knit Scarf
Cold Weather Ear Warmers
Hooded Warm Jacket
Retrieve
Tradational models
All items embedding
Top-K most similar items
Red Knit Scarf
Blue Ear Warmers
Hooded Warm Jacket
SigLIP2
Encoder
non-uniform
uniform
Non-Uniform Transformation
Residual Quantization
Inverse Transformation
Decoder
$\mathcal{L}_{\text{overall}} = \mathcal{L}_{\text{NUQ}} + \mathcal{L}_{\text{Sem}}$
Nearest
Codebook 1
Codebook 2
Codebook 3
Codebook 4
Encoder Output